Bank Telemarketing Success

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Introduction

Telemarketing is widely used in banking to promote financial products such as term deposits. However, traditional approaches often result in low response rates and inefficient targeting. Contacting uninterested customers increases operational cost and reduces campaign effectiveness.

Machine learning provides a data-driven approach to predict customer subscription likelihood. By identifying high-probability customers, banks can improve targeting and optimize resource allocation. This problem is challenging due to class imbalance, complex customer behavior, and feature dependencies.

Recent work shows that ensemble models improve prediction performance. Peter et al. (2025) demonstrate that stacking and boosting methods achieve strong results on imbalanced telemarketing data. Kaisar et al. (2024) propose an online ensemble framework to handle dynamic customer behavior, but highlight issues in retraining and computational cost. Bogireddy and Murari (2024) show that boosting-based models outperform traditional classifiers, while simpler models suffer from overfitting and minority class errors .

In this work, we evaluate multiple machine learning models, including Logistic Regression, KNN, Decision Tree, and Random Forest, alongside dimensionality reduction techniques. The goal is to improve predictive performance while addressing imbalance, overfitting, and practical deployment constraints.

Dataset

This project uses the Bank Telemarketing Campaign dataset from a Portuguese banking institution, containing 41,188 records across 20 features (4 main groups) and 1 binary target variable y, indicating whether a client subscribed to a term deposit or not.

The first issue that we identified is the missing values are not represented as NaN but encoded as the string "unknown" in categorical fields and as -1 in the certain binary fields. As can be seen in the Figure 1, the target variable y showed a clear class imbalance, which could bias the model toward predicting the majority class. In addition, the pdays variable used the value 999 to indicate that a customer had not been previously contacted, meaning it could not be interpreted directly as a normal numeric value. Moreover, numeric features campaign and previous showed significant right-skewness.

To address these issues, we performed some preprocessing steps. We dropped the duration column to avoid data leakage and removed duplicate rows. The pdays feature was split into two parts: its original values were set to NaN for unreached customers, and a new binary feature, had_previous_contact, was created to distinguish first-time from returning contacts. To reduce right skewness, campaign and previous features were transformed using log(1+x). In the preprocessing pipeline, categorical features were imputed with the most frequent value and one-hot encoded, while numeric features were median-imputed and standardised. Finally, SMOTE was applied to the training set after the train/test split to handle class imbalance.

Bar plots by target class were used to examine all categorical features. Based on Figure 2, higher subscription rates were observed among customers contacted by cellular phone, those without personal loans, and those with higher education levels. Some months, especially March, September, October, and December, showed relatively high subscription rates despite lower contact volume, suggesting possible seasonal campaign effects.

For numeric features, histograms and boxplots were used, while had_previous_contact was visualized with a bar plot. Based on Figure 3 and Figure 5, the campaign, previous and had_previous_contact features showed that most customers were contacted only 1 to 3 times and that most had no prior contact. Boxplots from Figure 4 also suggested that non-subscribers were associated with more extreme numbers of contact attempts, implying that excessive outreach may be ineffective. In addition, according to Figure 6, macroeconomic variables (emp.var.rate, euribor3m, and nr.employed) showed multimodal distributions, reflecting different economic phases, and subscribers were more concentrated in lower-value periods.

Based on the correlation matrix (Figure 7), the three macroeconomic features are highly intercorrelated, forming a multicollinearity. Therefore, in the modeling phase, we use PCA to mitigate the multicollinearity.

Methodology

Introduction of models

Three ensemble learning approaches – bagging, boosting, and stacking – have been proposed to improve prediction accuracy in bank telemarketing tasks (Peter et al., 2025). In this project, we apply bagging and boosting methods through Random Forest and XGBoost and compare their performance against base models, which are machine learning algorithms that make predictions independently, such as K Nearest Neighbours (KNN), Logistic Regression, and Decision Trees. Random Forest improves accuracy and reduces overfitting by training multiple decision trees on bootstrapped samples and combining their predictions (Peter et al., 2025). XGBoost builds models sequentially, where each subsequent model focuses on correcting the errors of its prior model, leading to improved predictive accuracy.

Data preparation and dimensionality reduction

Ratio between class "yes" to class "no" is approximately 8: 1 in the dataset. To deal with severe class imbalance, Synthetic Minority Over-sampling Technique (SMOTE) is used to generate synthetic samples for the minority class by interpolating between existing minority instances.

Before model training, Principal Component Analysis (PCA) was applied to reduce dimensionality and mitigate multicollinearity. After preprocessing, the dataset expanded from 12 to 60 features due to the use of one-hot encoding. Additionally, several socioeconomic features are highly correlated, thus motivating dimensionality reduction. PCA transforms correlated features into orthogonal components, removing redundancy while preserving key variance. In this project, 17 components were retained to explain 90% of total variance, significantly reducing feature space. This improves model stability and efficiency.

To assess model performance, we used Stratified K-fold Cross Validation (in this case, K = 5) to get more stable and reliable estimates of performance metrics like F1 score and AUC. This helps ensure that when we compare models, the evaluation is fair and not skewed by imbalanced folds.

Model	CV F1	CV F1 std	CV AUC	Test F1	Test Precision	Test Recall	Test AUC
Logistic Regression	0.712	0.004	0.7940	0.4345	0.3261	0.6511	0.7908
Decision Tree	0.701	0.003	0.7952	0.4447	0.3529	0.6011	0.7680
Random Forest	0.908	0.003	0.9645	0.4379	0.4087	0.4717	0.7579
XGBoost	0.855	0.005	0.932	0.430	0.371	0.512	0.754
KNN	0.887	0.003	0.930	0.359	0.273	0.526	0.699

Results & Discussions

For evaluation, we compare precision, recall, AUC and F1-score. Accuracy can be misleading for imbalanced datasets because it does not reflect how well the model performs on the minority class, which is the class we care about most. In this dataset, 89% of customers did not subscribe, meaning that a model that blindly predicts "No" for every customer achieves 89% accuracy while completely failing to identify a single potential subscriber. Since the goal of the telemarketing is to identify likely subscribers, a model that misses all of them has no practical value regardless of its accuracy score. For imbalanced datasets in the telemarketing context, recall is especially important as a low recall indicates that many actual subscribers are incorrectly predicted as non-subscribers, which can lead to missed business opportunities (Kaisar et al., 2024).

Overall, no model performed strongly on the minority class as seen in the consistently low F1 scores, highlighting the difficulty of the imbalanced classification task.

Across the models, Random Forest, XGBoost and KNN achieved the highest cross-validation (CV) F1 and AUC scores, indicating strong learning on training data. However, a significant drop in the F1 scores happen when these models are evaluated on the held-out test set, a result that suggests overfitting on minority samples generated by SMOTE. Similarly, test AUC decreases significantly across most models, except Logistic Regression which maintained almost identical scores (CV AUC = 0.7940, test AUC = 0.7908), indicating stability on unseen data.

Random Forest achieves the highest precision of 0.4087, but lowest recall of 0.4717. In comparison, Logistic regression attains a lower precision of 0.3261 but the highest recall of 0.6511. Hence, a closer comparison of model performance reveals a fundamental tradeoff between precision and recall. The reason for low precision is mainly due to the severe imbalance in the test data. When a model predicts "yes" on the test set, it's more likely to be wrong simply because "yes" instances are rare. Precision measures the accuracy of positive predictions, hence with so few actual "yes" cases, even a small number of false positives drags precision down. The Random Forest model is more conservative, only predicting subscriptions when confident resulting in fewer but more accurate positive predictions, On the other hand, Logistic Regression casts a wider net, catching more true subscribers at the cost of more false positives.

Given that the business cost of missing a subscriber outweighs the cost of making an unnecessary call, Logistic Regression is the most suitable model for deployment in this telemarketing context.

Appendix

Lists of figures

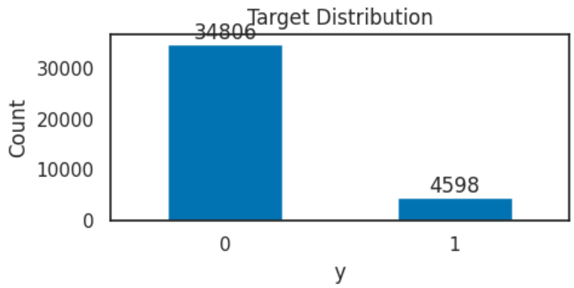
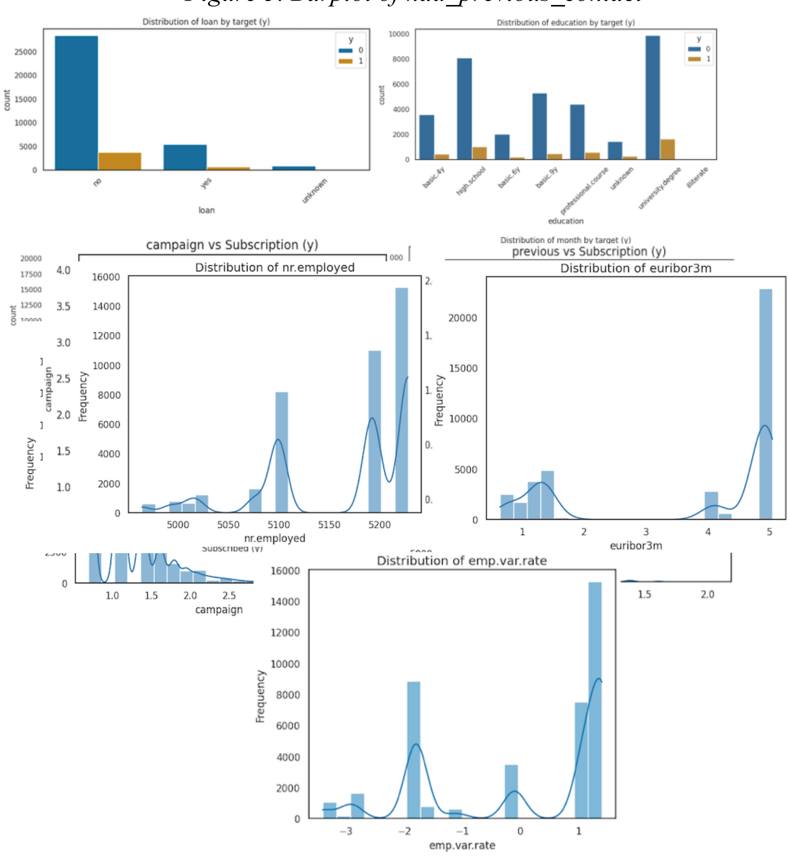


Figure 1. Target distribution

Figure 2. Barplots of education, loan, contact and month
Figure 3. Histograms of campaign and previous
Figure 4. Boxplots of campaign and previous

Figure 5. Barplot of had_previous_contact



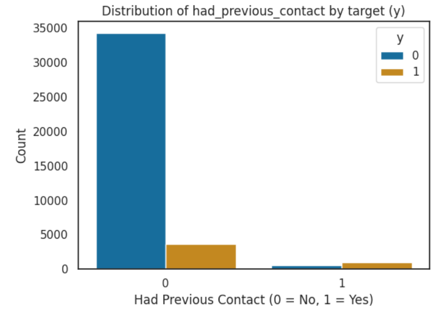


Figure 6. Histograms of emp.var.rate, euribor3m, and nr.employed

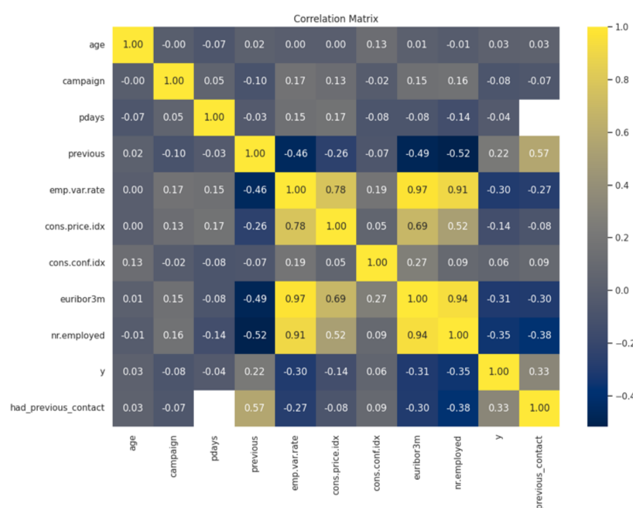
Figure 7. Correlation matrix of numeric features

AI Usage Declaration

AI tool used	How the tool was used in the work
ChatGPT	- Help with the initial direction and structure of project content. - Improve sentence clarity and wording. - Assist with code debugging and refinement.

References

Peter, M., Mofi, H., Likoko, S., Sabas, J., Mbura, R., & Mduma, N. (2025). Predicting customer subscription in bank telemarketing campaigns using ensemble learning



models. Machine Learning with Applications, 19, 100618. https://doi.org/10.1016/j.mlwa.2025.100618

Kaisar, S., Rashid, M. M., Chowdhury, A., Shafin, S. S., Kamruzzaman, J., & Diro, A. (2024). Enhancing telemarketing success using ensemble-based online machine learning. Big Data Mining and Analytics, 7(2), 294–314. https://doi.org/10.26599/BDMA.2023.9020041

Nashaat, M. (2024, December 29). Understanding SMOTE: A Powerful Technique for Handling Imbalanced Datasets. Medium. https://medium.com/@mohammednashaat29/understanding-smote-a-powerful-technique-for-handling-imbalanced-datasets-406ed6abda55

XGBoost developers. (n.d.). Get Started with XGBoost. https://xgboost.readthedocs.io/en/stable/get_started.html